

SPRINT - 11

Recruitment API Integration using Queueable Apex

1. Introduction

Today's work focused on integrating a Salesforce Application record with an external Recruitment API. The implementation uses Queueable Apex to perform an asynchronous HTTP callout, a Named Credential to define the external endpoint, an External Credential Principal and Permission Set to control access, and Application fields to store the integration result.

2. Important Definitions

Term	Definition
External API Integration	The process of allowing Salesforce to exchange data with an external application or service through an API.
HTTP Callout	A request sent from Salesforce Apex to an external web service. In this work, the request uses the HTTP POST method.
Queueable Apex	An asynchronous Apex mechanism used to execute work in the background. It is suitable here because the HTTP callout should run asynchronously.
Database.AllowsCallouts	An interface implemented by the Queueable class so that the asynchronous Apex job is permitted to make HTTP callouts.
Named Credential	A Salesforce configuration that represents the external service endpoint and its connection/authentication settings. The code refers to the configured endpoint through the callout name.
External Credential Principal	The principal associated with the external credential that defines access for the integration.
Permission Set	A Salesforce security mechanism used to grant additional permissions without changing the user's profile. The Recruitment API Access permission set was configured for the external credential principal.
JSON Serialization	Converting Apex data such as a Map into JSON text so it can be sent in an API request body.
JSON Deserialization	Converting a JSON response into Apex data structures so values returned by the API can be processed.
HTTP 201 Response	The API response shown in the execution logs indicating that the POST request was successfully accepted/created by the external service.
Integration Status	A field used to show whether the application synchronization is Pending, Sent, or Failed.
Integration Error	A field used to store an error message when the API callout or processing fails.

3. Application Object and Integration Fields

The Application object was checked in Object Manager. The Fields & Relationships page shows the fields used to capture application and integration information.

- Application Number – identifies the application record.
- Applied Date – stores the date on which the application was submitted.

- Student – identifies the related student.
- Job – identifies the related job.
- Status – stores the application status.
- Integration Status – stores the API synchronization state.
- External Candidate Id – stores the candidate identifier returned by the external API.
- Last Integration Attempt – records the most recent integration attempt.
- Integration Error – stores an error message when synchronization fails.

The screenshot shows the Salesforce Setup interface, specifically the 'Object Manager' section for the 'Application' object. The 'Fields & Relationships' tab is selected, displaying a list of 12 fields. The fields are sorted by label and include details such as field name, data type, controlling field, and indexed status. The interface includes a search bar, navigation tabs, and a sidebar with various setup options.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Application Number	Name	Text(80)		✓
Applied Date	Applied_Date__c	Date		
Created By	CreatedById	Lookup(User)		
External Candidate Id	External_Candidate_Id__c	Text(100)		
Integration Error	Integration_Error__c	Long Text Area(32768)		
Integration Status	Integration_Status__c	Picklist		
Job	Job__c	Lookup(Job)		✓
Last Integration Attempt	Last_Integration_Attempt__c	Date/Time		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Status	Status__c	Picklist		

4. Named Credential Configuration

A Named Credential named “Recruitment API Credential” was created for the external recruitment service. The screenshot shows the label, API name, and the configured authentication protocol.

Developer Edition

Setup > **NAMED CREDENTIALS**

Recruitment API Credential

Label: Recruitment API Credential Name: Recruitment_API_Credential

Authentication Protocol: No Authentication

Managed Package Access

Created By: Namespace

Related Named Credentials

You don't have any related named credentials.

Principals [New](#)

Developer Edition

Search Setup

Principals [New](#)

Sequence N...	Parameter Name	Authentication Status	Actions
1	Recruitment_API_Principal	Configured	New

Custom Headers [New](#)

You don't have any custom headers.

5. Queueable Apex Implementation

The integration logic is implemented in the CandidateSyncQueueable Apex class. The class implements Queueable and Database.AllowsCallouts. It receives the Application record Id, creates an HTTP request, prepares candidate data, serializes the data to JSON, sends the request, and reads the response.

5.1 Class Declaration and Request Setup

```
Developer Console - Google Chrome
curious-raccoon-5fz8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

1 public class CandidateSyncQueueable implements Queueable, Database.AllowsCallouts {
2
3     private Id applicationId;
4
5     public CandidateSyncQueueable(Id applicationId) {
6         this.applicationId = applicationId;
7     }
8
9     public void execute(QueueableContext context) {
10
11         HttpRequest request = new HttpRequest();
12
13         request.setEndpoint(
14             'callout:Recruitment_API/posts'
15         );
16
17         request.setMethod('POST');
18
19         request.setHeader(
20             'Content-Type',
21             'application/json'
22         );
23
24         Map<String, Object> candidateData =
25             new Map<String, Object>();
26
27         candidateData.put('applicationId', applicationId);
28         candidateData.put('studentId', 'STU1001');
29         candidateData.put('name', 'Pravallika');
30         candidateData.put('email', 'pravallika@example.com');
31         candidateData.put('branch', 'AIML');
```

5.2 Candidate Data and JSON Body

```
Developer Console - Google Chrome
curious-raccoon-5fz8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

27 candidateData.put('applicationId', applicationId);
28 candidateData.put('studentId', 'STU1001');
29 candidateData.put('name', 'Pravallika');
30 candidateData.put('email', 'pravallika@example.com');
31 candidateData.put('branch', 'AIML');
32 candidateData.put('cgpa', 8.5);
33 candidateData.put('jobId', 'JOB1001');
34 candidateData.put('company', 'ABC Technologies');
35 candidateData.put('role', 'Salesforce Developer');
36
37 request.setBody(
38     JSON.serialize(candidateData)
39 );
40
41 Http http = new Http();
42
43 try {
44
45     HttpResponse response =
46         http.send(request);
47
48     Integer statusCode =
49         response.getStatusCode();
50
51     System.debug(
52         'Response Status: ' + statusCode
53     );
54
55     System.debug(
56         'Response Body: ' +
57         response.getBody()
```

```
Developer Console - Google Chrome
curious-raccoon-5f1z8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

37 request.getBody()
38     JSON.serialize(candidateData)
39 );
40
41 Http http = new Http();
42
43 try {
44
45     HttpResponse response =
46         http.send(request);
47
48     Integer statusCode =
49         response.getStatusCode();
50
51     System.debug(
52         'Response Status: ' + statusCode
53     );
54
55     System.debug(
56         'Response Body: ' +
57         response.getBody()
58     );
59
60 } catch (Exception e) {
61
62     System.debug(
63         'Callout Error: ' +
64         e.getMessage()
65     );
66 }
67 }
68 }
```

5.3 Response Handling

The improved implementation processes the HTTP response. When the status code is 201, the response body is deserialized, Integration Status is set to Sent, the returned external id is stored, and Integration Error is cleared. For a non-success response, the record is marked Failed and the API status/body is stored as an error. A catch block handles exceptions and records the failure.

```
if (statusCode == 201) {
    Map<String, Object> responseData =
        (Map<String, Object>) JSON.deserializeUntyped(
            response.getBody()
        );

    applicationRecor.Integration_Status__c = 'Sent';

    if (responseData.containsKey('id')) {
        applicationRecord.External_Candidate_Id__c =
            String.valueOf(responseData.get('id'));
    }

    applicationRecord.Integration_Error__c = null;
} else {
    applicationRecord.Integration_Status__c = 'Failed';

    applicationRecord.Integration_Error__c =
```

```

'API returned status ' + statusCode +
': ' + response.getBody();
}

update applicationRecord;

catch (Exception e) {
    applicationRecord.Integration_Status__c = 'Failed';
    applicationRecord.Integration_Error__c = e.getMessage();
    applicationRecord.Last_Integration_Attempt__c = System.now();

    update applicationRecord;

    System.debug(
        'Callout Error: ' + e.getMessage()
    );
}

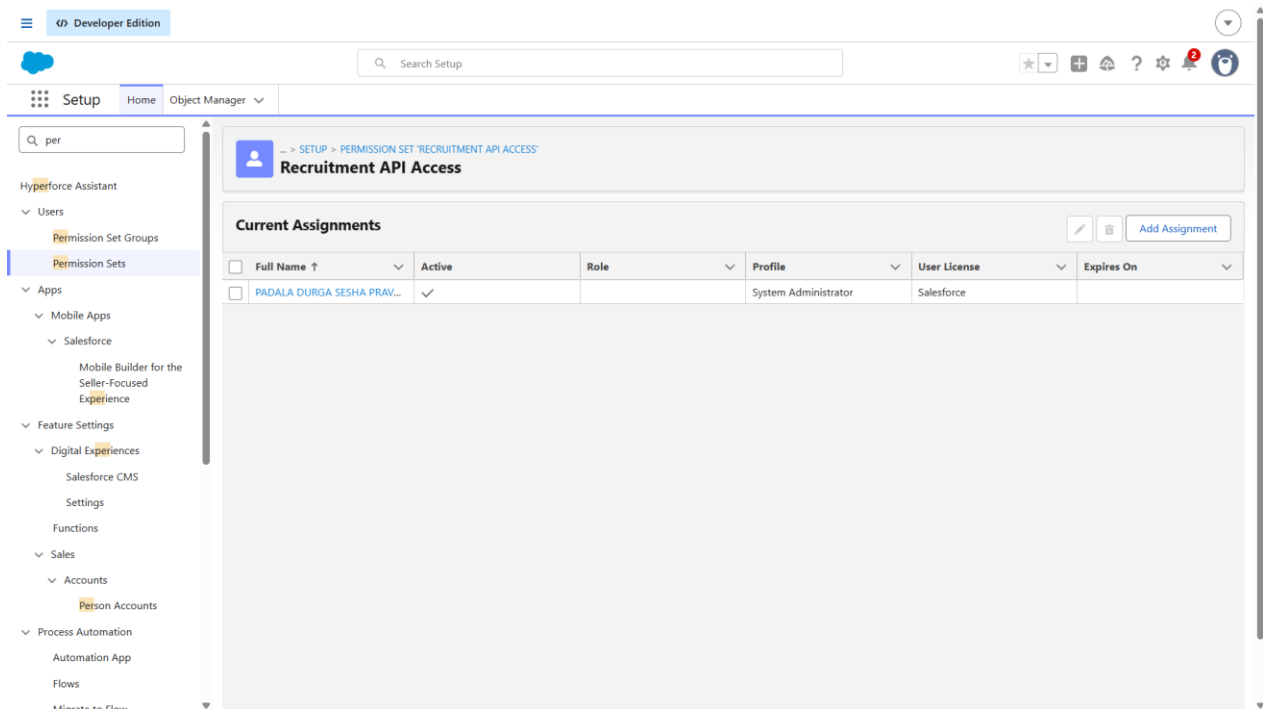
```

6. Permission Set and External Credential Access

The Recruitment API Access permission set was used to provide access to the external credential principal. The screenshots show External Credential Principal Access and the assignment to the Salesforce user.

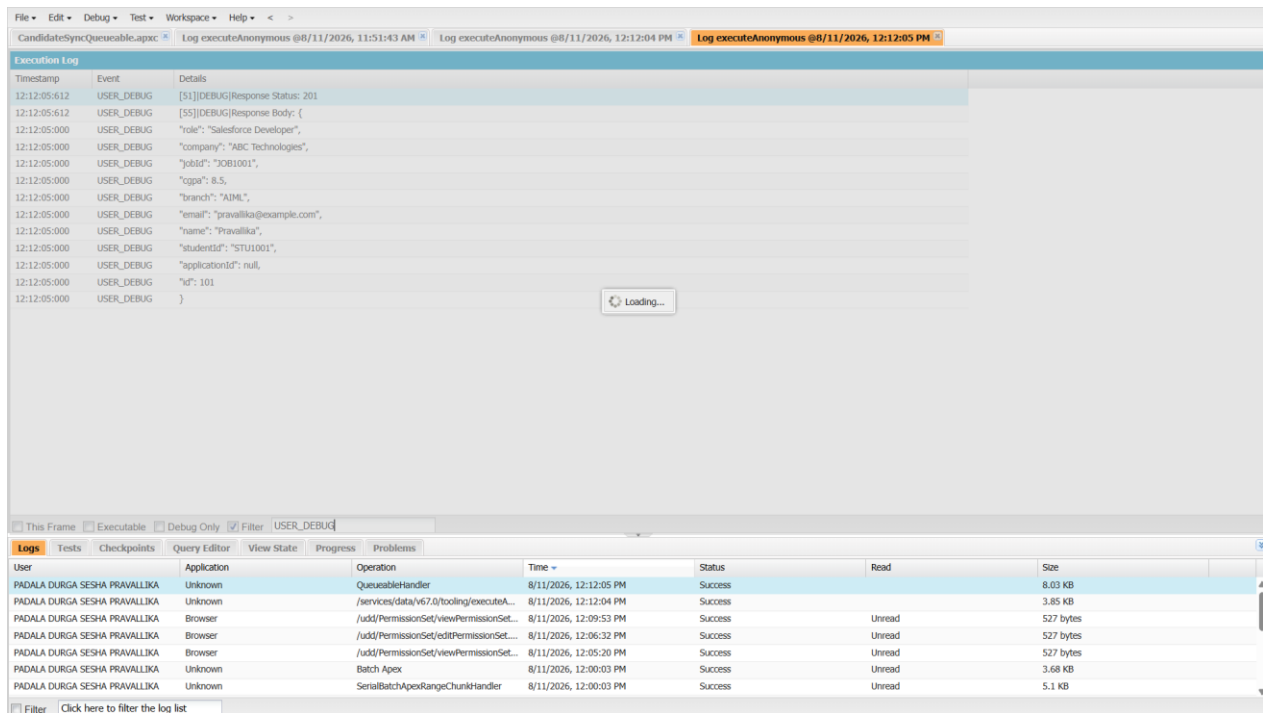
The screenshot displays the Salesforce Setup interface in Developer Edition. The left sidebar shows the navigation menu with 'Setup' selected. The main content area is titled 'Permission Sets' and shows the 'Recruitment API Access' permission set. The 'External Credential Principal Access' tab is selected, showing a list of external credential principals. The table below shows one entry: 'Recruitment_API_Credential - Recruitment_API_Principal' with the status 'Installed Package'.

External Credential Principal Name	Installed Package
Recruitment_API_Credential - Recruitment_API_Principal	Installed Package



7. Testing and Execution

The Queueable Apex was executed from the Developer Console. The execution logs show successful QueueableHandler operations and the API response. The API returned HTTP status 201 and an id value of 101 in the response.



Developer Edition

Search...

Playground Starter Welcome Install a Package Get Your Login Credentials * APP-005 | Application

Application APP-005

New Contact Edit New Opportunity

Related Details

Application Number	APP-005	Owner	PADALA DURGA SSHA PRAVALLIKA
Student	SOWJANYA Updated		
Job	INFOSYS		
Status	Applied		
Applied Date	11/08/2026		
Integration Status	Sent		
External Candidate Id	101		
Last Integration Attempt	11/08/2026, 12:20 pm		
Integration Error			
Created By	PADALA DURGA SSHA PRAVALLIKA, 11/08/2026, 12:17 pm	Last Modified By	PADALA DURGA SSHA PRAVALLIKA, 11/08/2026, 12:20 pm

9. Repeated Verification and Final Run

The integration was executed and verified again later in the day. The application record continued to show Integration Status = Sent and External Candidate Id = 101. The Developer Console logs again recorded HTTP 201 and a successful QueueableHandler execution.

Developer Edition

Search...

Playground Starter Welcome Install a Package Get Your Login Credentials * APP-005 | Application

Application APP-005

New Contact Edit New Opportunity

Related Details

Application Number	APP-005	Owner	PADALA DURGA SSHA PRAVALLIKA
Student	SOWJANYA Updated		
Job	INFOSYS		
Status	Selected		
Applied Date	11/08/2026		
Integration Status	Sent		
External Candidate Id	101		
Last Integration Attempt	11/08/2026, 12:20 pm		
Integration Error			
Created By	PADALA DURGA SSHA PRAVALLIKA, 11/08/2026, 12:17 pm	Last Modified By	PADALA DURGA SSHA PRAVALLIKA, 11/08/2026, 2:11 pm

File • Edit • Debug • Test • Workspace • Help • < >

Log executeAnonymous @8/11/2026, 12:20:33 PMLog executeAnonymous @8/11/2026, 12:20:34 PMApplicationTrigger.apxtLog executeAnonymous @8/11/2026, 2:24:18 PMLog executeAnonymous @8/11/2026, 2:27:03 PM

Execution Log

Timestamp	Event	Details
14:27:03:000	USER_INFO	[EXTERNAL]005hg00001eh2U[2]3pa1a4281@curious-raccoon-5fz8.com[GMT+05:30] India Standard Time (Asia/Kolkata)[GMT+05:30]
14:27:03:983	USER_DEBUG	[84][DEBUG]Response Status: 201
14:27:03:983	USER_DEBUG	[88][DEBUG]Response Body: {
14:27:03:000	USER_DEBUG	"appliedDate": "2026-08-11",
14:27:03:000	USER_DEBUG	"jobId": "a01hg00000sp2QvAAT",
14:27:03:000	USER_DEBUG	"studentId": "a00hg00000F1Q3HAA3",
14:27:03:000	USER_DEBUG	"applicationNumber": "APP-005",
14:27:03:000	USER_DEBUG	"applicationId": "a02hg000001JW744AG",
14:27:03:000	USER_DEBUG	"id": 101
14:27:03:000	USER_DEBUG	}

Loading...

This Frame Executable Debug Only Filter USER

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SETHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:27:03 PM	Success	Unread	5.51 KB
PADALA DURGA SETHA PRAVALLIKA	Unknown	QueueableHandler	8/11/2026, 2:27:03 PM	Success		13.12 KB
PADALA DURGA SETHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:26:26 PM	Success	Unread	3.93 KB
PADALA DURGA SETHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:24:18 PM	Success		3.92 KB

Filter Click here to filter the log list

Developer Edition

Q Search...

Playground StarterWelcomeInstall a PackageGet Your Login CredentialsAPP-005 | Application

Application

APP-005

New ContactEditNew Opportunity

RelatedDetails

Application Number

APP-005

Student

SOWJANYA Updated

Job

INFOSYS

Status

Selected

Applied Date

11/08/2026

Integration Status

Sent

External Candidate Id

101

Last Integration Attempt

11/08/2026, 2:27 pm

Integration Error

Created By

PADALA DURGA SETHA PRAVALLIKA, 11/08/2026, 12:17 pm

Last Modified By

PADALA DURGA SETHA PRAVALLIKA, 11/08/2026, 2:30 pm

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SESHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:50:44 PM	Success	Unread	3.93 KB
PADALA DURGA SESHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:49:24 PM	Success	Unread	5.54 KB
PADALA DURGA SESHA PRAVALLIKA	Unknown	QueueableHandler	8/11/2026, 2:49:24 PM	Success		13.07 KB
PADALA DURGA SESHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:37:09 PM	Success	Unread	3.93 KB
PADALA DURGA SESHA PRAVALLIKA	Browser	/aura	8/11/2026, 2:30:47 PM	Success	Unread	3.93 KB
PADALA DURGA SESHA PRAVALLIKA	Unknown	QueueableHandler	8/11/2026, 2:27:03 PM	Success		13.12 KB

10. Overall Integration Workflow

1. An Application record is created or updated in Salesforce.
2. The application information is passed to the asynchronous integration process.
3. CandidateSyncQueueable executes in the background and is allowed to make callouts.
4. The Apex class creates an HTTP POST request to the Recruitment API through the configured Named Credential.
5. Candidate information is stored in a Map and serialized into JSON.
6. Salesforce sends the JSON payload to the external API.
7. The API returns a response; the successful runs shown in the logs return HTTP 201 and an external id of 101.
8. Salesforce stores the returned external candidate id and updates Integration Status to Sent.
9. The Last Integration Attempt field records the time of the synchronization.
10. If an error occurs, Integration Status is set to Failed and the Integration Error field records the error message.

11. Problems Encountered and Resolution

During implementation, the integration was refined so that the Salesforce Application record does not only send the data but also records the result of the API operation. Response handling was added for successful status 201, non-success status codes, and exceptions. This makes the integration easier to monitor and debug.

- Success handling: set Integration Status to Sent and store the returned external id.
- Failure handling: set Integration Status to Failed and save the API response as an integration error.
- Exception handling: capture the exception message and update the application record.
- Monitoring: use Developer Console debug logs to verify response status and response body.